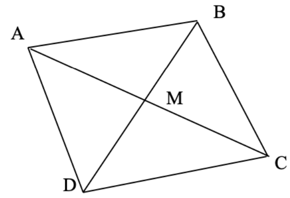
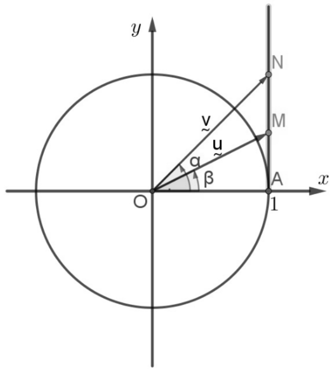
Vector WS 3

1. The point A has position vector $\hat{a} = 2\hat{\imath} + 2\hat{\jmath} + \hat{k}$ and point B has position vector $\hat{b} = \hat{\imath} + \hat{\jmath} - 4\hat{k}$, relative to an origin O.
   a. Find the position vector of point C, with position vector $\hat{c}$ given by
      $\hat{c} = \hat{a} + \hat{b}$.
   b. Show that $OACB$ is a rectangle.
   c. Find the exact area of $OACB$.



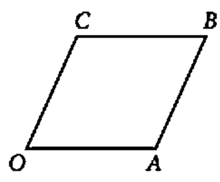
2. Consider the quadrilateral $ABCD$ shown. The diagonals AC and BD bisect each other at M. Copy the diagram. Use the vector to prove that $\overrightarrow{AB} = \overrightarrow{DC}$.
   [Hint: for convenience, let $\overrightarrow{AM} = p$ and $\overrightarrow{MB} = q$.]

3. A line passes through the points $A(1, 3, -2)$ and $B(2, -1, 5)$.
   a. Show that the vector question of the line AB is given by:
      $\hat{r} = (\hat{\imath} + 3\hat{\jmath} - 2\hat{k}) + \lambda_1(\hat{\imath} - 4\hat{\jmath} + 7\hat{k}),\ \lambda_1 \in \mathbb{R}$.
   b. Determine if the point $C(3, 4, 9)$ lies on the line.
   c. Consider a line with parametric equations $x = 1 - \lambda_2, y = 2 + 3\lambda_2$ and $z = -1 + \lambda_2$. Assuming the line is neither parallel or perpendicular to AB, determine whether the lines intersect or are skew.



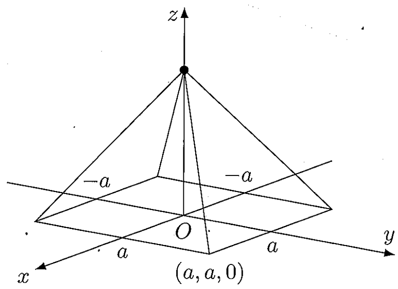
4. Consider the points $M = (1, \tan\beta)$ and $N = (1, \tan\alpha)$, where $\alpha = \angle AON$ and $\beta = \angle AOM$ as shown on the diagram.
   a. Find a point P such that $OMPN$ is a parallelogram
   b. Show that the area of a parallelogram with adjacent sides $\hat{u}$ and $\hat{v}$ is
      $$\frac{|\hat{v}|\hat{u}|^2 - (\hat{v}.\hat{u})\hat{u}|}{|\hat{u}|}.$$
   c. Hence, find the area of $OMPN$.

5. Consider the vector lines $v_1 = \begin{bmatrix} 8 \\ 16 \\ 6 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 14 \\ 12 \\ 10 \end{bmatrix} + \lambda \begin{bmatrix} 12 \\ 8 \\ 10 \end{bmatrix}$.
   a. Find the point of intersections of the lines
   b. Find the acute angle between the lines (to the nearest degree).
6. Find the equation of the sphere with centre $(4, 5, -2)$ that passes through the point $(1, 0, 0)$.
7. Show that the line through the points $(1, -1, 1)$ and $(5, 3, 3)$ is perpendicular to the line through the points $(1, -1, 2)$ and $(4, -4, 6)$.
8. For the curves whose vector equations are given, find the parametric equation and the Cartesian equation.
   a. $\vec{r} = (2\sin t)\vec{\imath} + (4\cos t)\vec{\jmath}, t \in \mathbb{R}$
   b. $\vec{r} = \left(t - \frac{1}{t}\right)\vec{\imath} + \left(t^2 + \frac{1}{t^2}\right)\vec{\jmath}, t \neq 0$

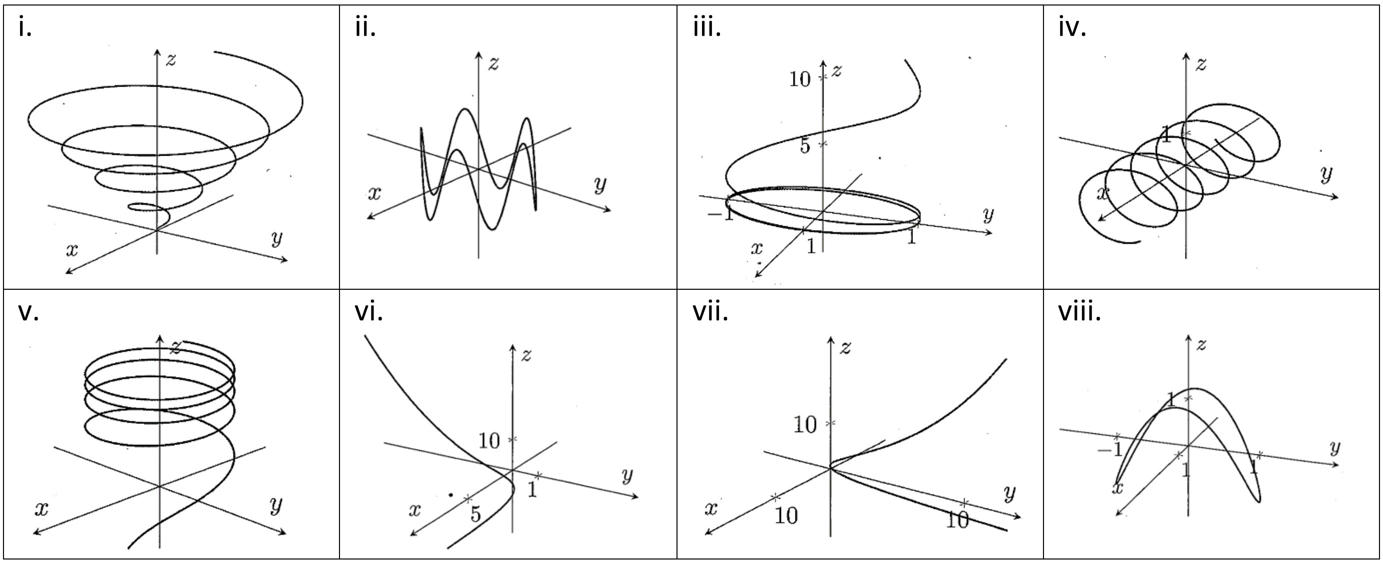


9. In the rhombus $OABC$, shown in the diagram, $\overrightarrow{OA} = m\vec{\imath}$ and $\overrightarrow{OC} = \vec{\imath} + \vec{\jmath} + \vec{k}$, where $m \in \mathbb{R}$.
   a. Find the value of m.
   b. Show that the diagonals of $OABC$ are perpendicular.

10. The diagram below shows a square pyramid where the side faces are equilateral triangles. Let the coordinates of the square base be $(a, a, 0), (a, -a, 0), (-a, -a, 0)$.
    a. Find the coordinates of the tip of the vertex in terms of a
    b. Find the angle of inclination between any of the side faces and the square base.



11. The diagrams below show sketches of eight parametrically defined curves.



Match the curves to the appropriate set of parametrisations below.
    a. $\vec{r}(t) = (\cos t)\vec{\imath} + (\sin t)\vec{\jmath} + (e^{-0.5t})\,\vec{k}$
    b. $\vec{r}(t) = (t\cos 5t)\vec{\imath} + (t\sin 5t)\vec{\jmath} + (t)\,\vec{k}$
    c. $\vec{r}(t) = (t)\vec{\jmath} + (t^2)\vec{\jmath} + (t^3)\,\vec{k}$
    d. $\vec{r}(t) = (\cos t)\vec{\imath} + (\sin t)\vec{\jmath} + (\cos 2t)\,\vec{k}$
    e. $\vec{r}(t) = \cos(t)\,\vec{\imath} + \sin(t)\ \vec{\jmath} + (\ln t)\,\vec{k}$
    f. $\vec{r}(t) = (e^t)\vec{\imath} + (t)\vec{\jmath} + (t^2)\,\vec{k}$
    g. $\vec{r}(t) = (t)\vec{\imath} + (\cos 6t)\vec{\jmath} + (\sin 6t)\,\vec{k}$
    h. $\vec{r}(t) = (\cos t)\vec{\imath} + (\sin t)\vec{\jmath} + (\sin 5t)\,\vec{k}$

12. Let $\vec{a}$ and $\vec{b}$ be the position vectors of two fixed points A and B in space, and let $\vec{v}$ be a variable position vector that satisfies $(\vec{v} - \vec{a}) \cdot (\vec{v} - \vec{b}) = 0$.
    a. Show that $\left|\vec{v} - \frac{\vec{a}+\vec{b}}{2}\right| = \frac{1}{2}\left|\vec{a} - \vec{b}\right|$.
    b. Geometrically, what does the position vector $\vec{v}$ represent, and find the main features of it.
    c. What is the geometric significance of the above results?

Answer

1. $A = 9\sqrt{2}\,u^2$
3. b. No; c. do not intersect skew
4. a. $P(2, \tan\alpha + \tan\beta)$; c. $\tan\alpha - \tan\beta\ \ (\alpha > \beta)$
5. a. $\begin{bmatrix}2\\4\\0\end{bmatrix}$; b. 28°
6. $(x-4)^2 + (y-5)^2 + (z+2)^2 = 38$
8. a. Parametric: $x = 2\sin t, y = 4\cos t, t \in \mathbb{R}$, Cartesian: $\frac{x^2}{4} + \frac{y^2}{16} = 1$; b. Parametric: $x = t - \frac{1}{t}, y = t^2 + \frac{1}{t^2}, t \neq 0$, Cartesian: $y = x^2 + 2$
9. a. $m = \sqrt{3}$
10. a. $(0, 0, a\sqrt{2})$; b. $\tan^{-1}(\sqrt{2})$
11. a. iii; b. i; c. vii; d. viii; e. v; f. vi; g. iv; h. ii
12. b. circle with diameter AB; c. It proves that the set of all points that form perpendicular vectors with respect to two fixed points is a circle. In other words it proves the familiar 'angle in a semi-circle' property otherwise known as *Thales' Theorem*